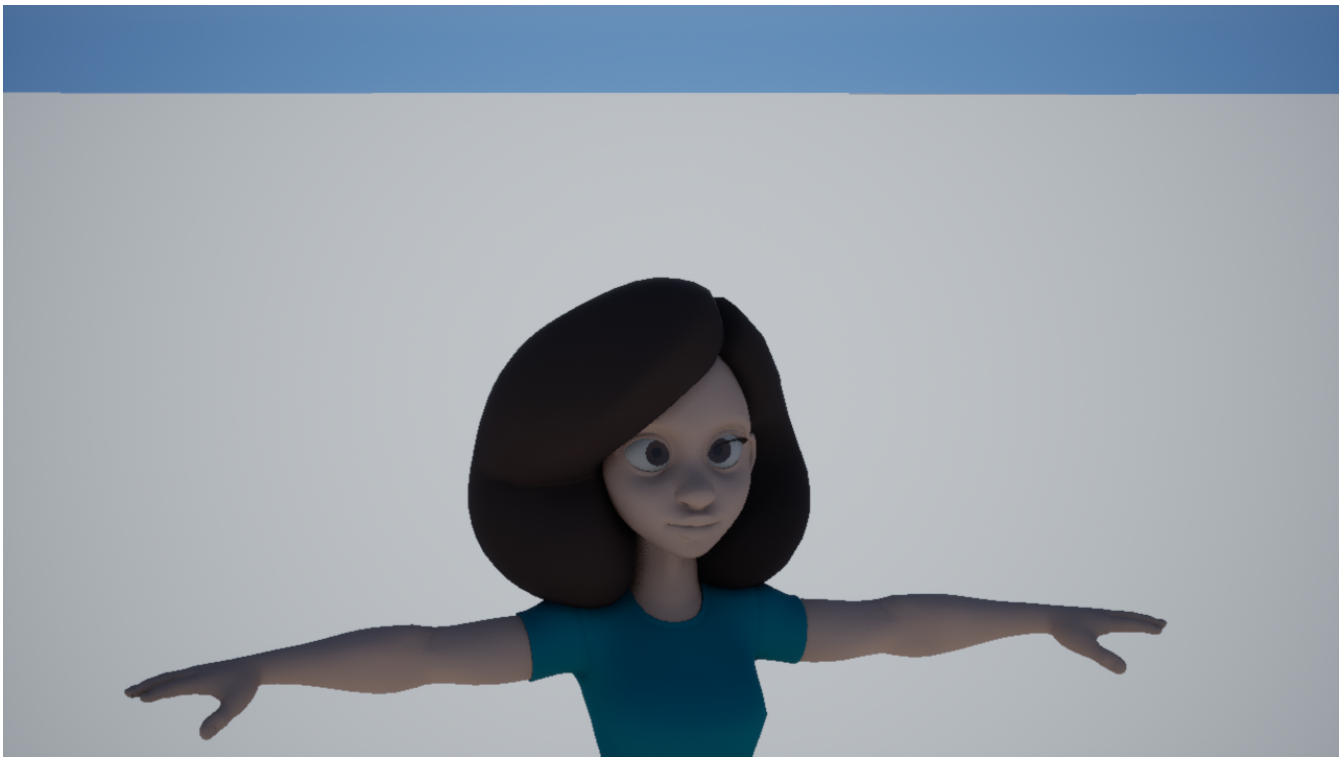
FILM 304 CLASS REVIEW COPY

Humber Film 304 Production Bible

Toon materials, asset names and Dana handoff



Dana material and lighting preview. This static pose does not verify body or face deformation.

Prepared for the Film 304 group review | 16 October 2026

PROJECT SETUP

Project snapshot

This guide records the Humber Toon Spine material setup, the fixed deep-night sky preset, and Dana's current export status. Use it to find the right material, adjust its controls, follow asset names, and see what still needs a fresh render or rig export.

The build rule
Material assets are generated from the Python builders in this repository. Change the builder, rebuild the asset, then read it back in Unreal. An editor-only graph edit will be lost on the next rebuild.

Step	What the artist chooses	Where it lives
1. Choose a family	Skin, hair, face, glass, cloth, ground, or sky.	M_* master
2. Set the surface	Tint, roughness, line weight, or a surface pattern.	MI_* instance
3. Set light bands	Choose the light and shadow response for the subject.	TP_* profile
4. Review the shot	Check the look under the shot's lighting and grade.	Rendered still and notes

Working rules

- Assign one person to each Unreal binary while it is open for editing. Put look changes in material instances.
- Change the builder that makes an asset, rebuild it, and read back the result. Save a still and the build report.
- Use the SM6 workstation to judge final shader output. The Oct 8 laptop report records Substrate surfaces rendering black.

Open reviews
The six film functions and fixed night sky preset compiled after a fresh editor load on Oct 7. The sky still needs review in SH010 and SH020. Dana needs a new Maya export before body or face animation can be approved.

MATERIAL ASSIGNMENTS

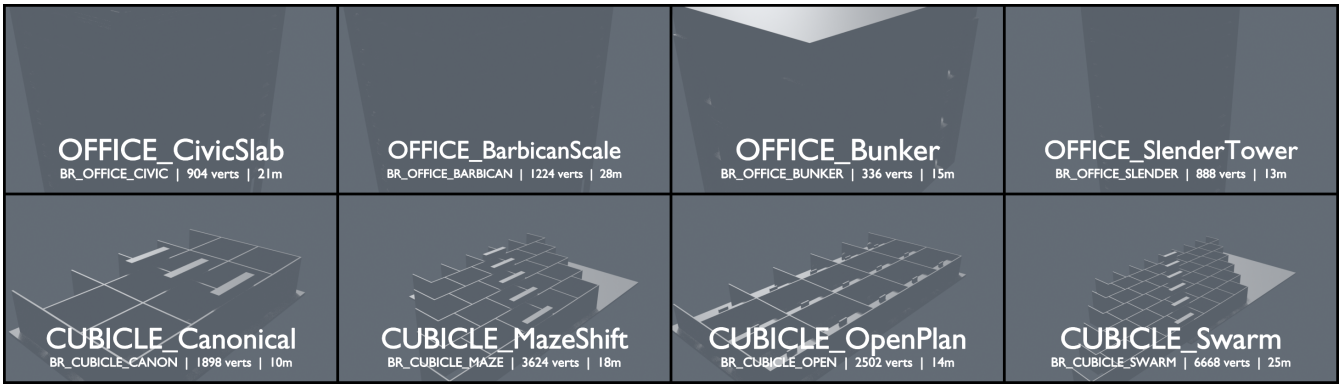
Material setup

Use a master for the surface family, a toon profile for its light and shadow bands, and a material instance for color and surface detail. Apply outlines and the final film grade separately.

Surface	Start with	Use it for
Shared surfaces	M_Master_Toon_Universal + MI_Toon_*	Office props, masonry, and surfaces that share the main band response.
Character body	M_Master_Toon_Character	Clothing and broad body shading.
Face and hair	M_Master_Toon_Face / M_Master_Toon_Hair	Face shadow shape, scalp, and hair response.
Eyes and cornea	MI_DanaFilm_EyeWhite / MI_DanaFilm_Iris / MI_DanaFilm_EyeInterior / MI_DanaFilm_Cornea	Separate controls for the sclera, iris, eye interior, and cornea.
Landscape	M_Master_Toon_Landscape	Ground and broad terrain variation.
Sky	M_Master_Toon_Sky + MI_Toon_Sky_DeepNight	One fixed deep-night look.
Final grade	M_Master_Toon_PostComposite	Apply the shared grade before any shot-specific tint.
Line work	M_Outline_InvertedHull + MI_Outline_Thin / MI_Outline_Heavy	Use thin outlines on characters and heavier outlines on architecture.

Light and outline guidelines

- Keep the darkest profile band above pure black so shadowed forms retain their shape.
- Set light and shadow bands in the toon profile. Set color and surface detail in the material instance.
- Apply one final grade in the post-composite material. Add a shot tint only when the shot calls for it.



Office and cubicle model families from the class staging scaffold.

REUSABLE CONTROLS

Film controls and deep-night sky

These six functions start at neutral values. Raise a control when a shot needs the effect, then check it under the final lighting.

Function	Useful for	Controls and starting values
MF_NormalAdjust	Soften or sharpen toon bands across curved forms.	NormalStrength 1.0 is identity; below 1 softens, above 1 tightens.
MF_SurfaceWear	Small worn patches and crack marks on walls or floors.	WearStrength 0; CrackStrength 0. WearScale 0.02 is about a 50 cm tile. CrackWidth 0.01-0.08.
MF_DF_ContactBlend	Dirt and roughness close to contact points.	DFContactStrength 0. Raise it to add contact dirt; tint and roughness stay unchanged at zero.
MF_Impasto	Raised, directional paint strokes.	ImpastoStrength 0. BrushScale 0.02-0.09; 0.045 gives broad strokes.
MF_ClothWindDrape	Subtle motion in cloth and stage fabric.	WindStrength 0-1 cm; 0.05-0.15 for a drift, 0.2-0.35 for a walk breeze. Speed 0.2-1.5 cycles/sec.
MF_FilmGrade	A small neutral finish across the image.	Contrast 1; Warmth 0; Saturation 1; Lift 0. Keep GradeTint at the shot tint.

Deep-night preset
MI_Toon_Sky_DeepNight is an instance of M_Master_Toon_Sky. It has an indigo-black zenith, a city-glow horizon, four value bands, and sparse stars. The preset is fixed and has no time-of-day controls.

Leave iridescence, sparkle, vein glow, and radial rings out of the film materials. Add another function only when a shot needs a specific effect.

FILE AND NAMING RULES

Asset names and handoff

Inside Unreal, use the prefixes in Docs/CONTENT_CONVENTIONS.md. Use role prefixes only in the Humber staging folders. A staging prefix identifies the incoming source file; an Unreal prefix identifies the asset in the project.

Unreal asset	Prefix	Example
Master material	M_	M_Master_Toon_Universal
Material instance	MI_	MI_Toon_Stone
Material function	MF_	MF_RampLUT
Toon profile	TP_	TP_Hero
Texture	T_	T_Concrete_N
Static mesh	SM_	SM_Brutalist_Lot_A
Blueprint	BP_	BP_Brutalist_Kit
Level / sequence	L_ / LS_	L_Toon_Lookdev / LS_Shot010

Staging folder examples

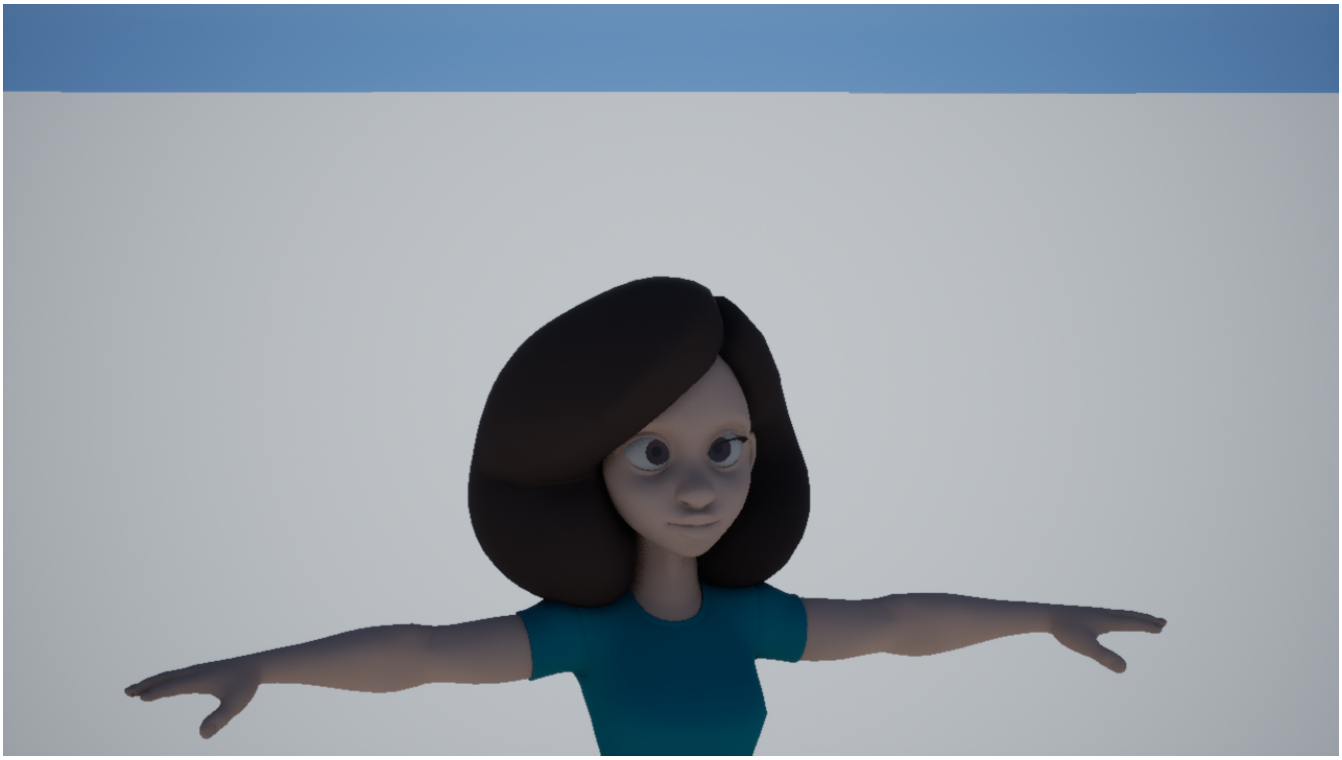
Incoming work	Folder and prefix
Character model	02_Assets/Models/Characters/CH_*
Character rig	02_Assets/Rigs/Character/RIG_CH_*
Environment model	02_Assets/Models/Environments/EN_*
Prop model	02_Assets/Models/Props/PR_*
Material work	02_Assets/Materials/M_*, MI_*, TP_*, T_*

- For textures, use _D for diffuse, _N for normal, _R for roughness, and _M for masks.
- Use a descriptive word for a real variant, such as _Thin or _Heavy. Avoid names like Stone2 and final_final.
- The staging guide uses _v01 for versioned source files. Keep Unreal asset names stable once other files reference them.
- The UE naming guide does not specify a SkeletalMesh prefix. Agree on one with the group before Dana's final import.

Before import
Confirm the owner, source path, units, up axis, material slots, and skin weights. Record the intended Unreal name and keep the source FBX beside its import notes.

DANA HANDOFF

Dana rig and lash findings



Daylight preview. This is a static pose; the upper lash line is faint, and the image does not verify body or face deformation.

Area	Audit result
Material stage	The preview build has one Dana actor and 11 assigned material slots.
Body deformation	The source FBX body has no skin cluster, so this export cannot prove body deformation.
Upper/lower lashes	Each side has 39 vertices and 48 triangles. All lash vertices are weighted to FaceUpperRegion_M.
Visibility	The intended lash meshes and their FaceDeformationSystem parent are hidden in the export.
Squint meshes	The alternate squint pair has no skin weights. Exclude it from the render export.
Material slot	The lashes use danaHair_mat, which maps to the hair instance. The missing lash line is not a material assignment problem.

Maya export still needed
Turn on the intended upper/lower lash pair. Export the body with skin weights and include the face/head bones on the same skeleton as the body. Keep the lash weights on FaceUpperRegion_M. Test blink, closed eyes, look-up, and squint before replacing the Unreal preview.

The filtered FBX kept the lashes but still had no body skin cluster. Use it for diagnosis only; it is not a production rig export.

OFFICE SPIDER SHOT PLAN

Materials by shot range

This table follows the Office Spider shot plan. Check the storyboard and confirm changes with the shot owner before changing a material look.

Shot range	Material work	Status in the handoff notes
SH010, SH030	Brutalist exterior, landscape, stone, foliage, sky, and outline weight.	Exterior kit is staged. Deep-night sky still needs its SH010/SH020 review.
SH020-SH060	Cubicle plastic, carpet, laminate, paper, steel, screen, face, hair, and glass.	Interior pieces are staged. Character and spider rig binds remain a dependency.
SH070-SH120	Kitchen and hallway surfaces, glass, coffee props, spider and character materials.	Work is deferred until the required rigs and set build-out are ready.

Prop materials

- Paper uses a matte finish and paper grain. Donut boxes use the cardboard surface.
- The coffee machine uses brushed metal; VCT floor grout and carpet loops should still read at the final shot scale.
- Frosted glass belongs on the door glazing. Keep outlines thinner on characters and heavier on architecture.

Review materials in the shot
A green build report confirms the graph loaded and its parameters exist. The lighting, bands, outlines, and textures still need visual review in the shot.

REVIEW

Checks still to complete

Gate	Evidence on record	Still needed
Material build	Six film utilities and the static sky preset were built through Python; the Oct 7 fresh-process compile record was clean.	Refresh the gate after any builder changes.
Look review	Dana day and night stills exist; the capture report records three fresh images.	Judge final Substrate materials on the SM6 workstation. One Oct 13 render-queue attempt reported a stale image.
Dana rig	FBX anatomy, lash filter, and pose-clean reports are saved.	Maya source export with body skin weights and a working head/face chain.
Sky preset	Static parameters and preset are present in the spine.	View it in SH010 and SH020, then approve or adjust it.

Next steps

- Re-export Dana from Maya with one body skinCluster, visible upper/lower lashes, and the final face/head deformation chain.
- Import the export into a clean preview, check one character only, and read back every material slot.
- Refresh the stills on the SM6 workstation and review the sky preset in SH010 and SH020.
- After those reviews, stage the next shot range when the kitchen, hallway, and remaining rigs are ready.

Sources and records

Docs/CONTENT_CONVENTIONS.md; Docs/GROUP_WORKFLOW.md; Humber_FinalYear_Prep/GROUP_STAGING_GUIDE.md; Docs/TOON_SPINE.md; Docs/MELODIA_TOON_CONVERGENCE.md; Docs/HANDOFF_2026-10-08_NEXT_PHASE.md; Docs/OFFICE_SPIDER_SHOT_PLAN.md; Showcase/Dana/README.md; Saved/Audit/dana_fbx_raw_anatomy.json; Saved/Audit/dana_fbx_geometry_filter_Dana_Film_EyelashFiltered.json; Saved/Audit/dana_fbx_deform_prune_Dana_Film_EyelashPreview.json; Saved/Audit/dana_refined_uds_build.json; Saved/Audit/dana_rig_preview_capture.json; Saved/Audit/dana_refined_uds_render_results.json.